

Statistical Learning: Classification III – Linear Discriminant Analysis

Course of Statistical Learning
Gareth et al.. An Introduction to Statistical Learning, Ch. 4.4

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Linear Discriminant Analysis

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- **Logistic regression** involves directly modelling $Pr(Y = k | X = x)$ using the **logistic function** for the case of two response classes: we model the conditional distribution of the response Y , given the predictors X_i with $1 \leq i \leq p$
- But we can consider an **alternative and less direct approach** to **estimating these probabilities**
- We **model the distribution of the predictors X separately in each of the response classes** (i.e. given Y), and then use **Bayes' theorem** to flip these around into estimates for $Pr(Y = k | X = x)$
- When these distributions are assumed to be **normal**, it turns out that the model is very similar to **logistic regression**

Why use LDA?

Why do we need another method, when we have logistic regression? There are several **reasons**:

- When the **classes are well-separated**, the parameter estimates for the logistic regression model are surprisingly **unstable**. Linear discriminant analysis does not suffer from this problem.
- If n is **small** and the distribution of the predictors X is approximately **normal** in each of the classes, the linear discriminant model is again **more stable** than the logistic regression model.
- As mentioned, linear discriminant analysis is **popular when we have more than two response classes**.

Bayes' Theorem

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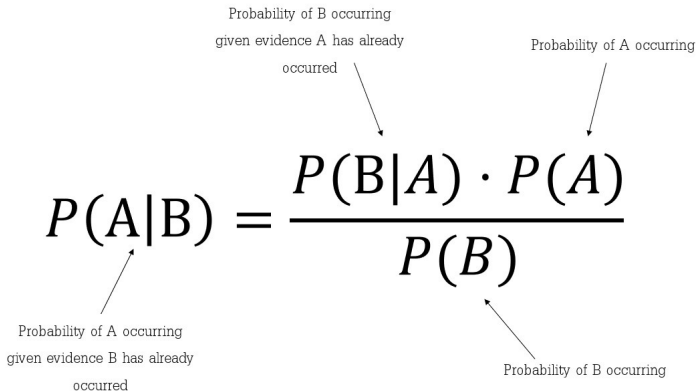


Diagram illustrating Bayes' Theorem with annotations:

Annotations:

- Probability of B occurring given evidence A has already occurred (points to $P(B|A)$)
- Probability of A occurring (points to $P(A)$)
- Probability of A occurring given evidence B has already occurred (points to $P(A|B)$)
- Probability of B occurring (points to $P(B)$)

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

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Using Bayes' Theorem for Classification

- Suppose that we wish to classify an observation into one of **K classes**, where $K \geq 2$. In other words, the categorical variable Y can take on K possible distinct and unordered values.
- Let π_k represent the **prior probability** that a randomly chosen observation comes from the k^{th} class; this is the probability that a given observation is associated with the k^{th} category of the response variable Y .
- Let $f_k(X) = Pr(X = x | Y = k)$ denote the **density function** of X for an observation that comes from the k^{th} class. $f_k(x)$ is relatively **large** if there is a **high probability** that an observation in the k^{th} class has $X \approx x$, and $f_k(x)$ is **small** if it is very **unlikely** that an observation in the k^{th} class has $X \approx x$.

Bayes' Theorem

Then **Bayes' theorem** states that

$$p_k(x) = \Pr(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{j=1}^K \pi_j f_j(x)} \quad (1)$$

- This suggests that instead of directly computing $p_k(X)$, we can simply plug in estimates of π_k and $f_k(X)$ into Equation (1)
- In general, **estimating** π_k **is easy** if we have a random sample of Y from the population: we simply compute the **fraction of the training observations** that belong to the k^{th} class
- Estimating $f_k(X)$ tends instead to be **more challenging**, unless we assume some simple **forms** for these densities

The idea behind LDA

- We refer to $p_k(x)$ as the **posterior probability** that an observation $X = x$ belongs to the k^{th} class. That is, it is the **probability that the observation belongs to the k^{th} class, given the predictor value for that observation**
- We know that the **Bayes classifier**, which classifies an observation to the class for which $p_k(X)$ is the largest, has the **lowest possible error rate** over all classifiers
- Therefore, if we can find a way to **estimate $f_k(X)$** , then we can develop a classifier that **approximates the Bayes classifier**

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Density of a Gaussian variable

- Assume that $p = 1$ that is, we have only **one predictor**. We would like to obtain an **estimate for** $f_k(x)$ that we can plug into Eq. (1) **to estimate** $p_k(x)$
- We will then **classify an observation** to the class for which $p_k(x)$ **is the greatest**. To estimate $f_k(x)$, we will first make some **assumptions about its form**, by requiring that $f_k(x)$ is **normal** or **Gaussian**
- With $p = 1$, the **normal density** takes the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right) \quad (2)$$

where μ_k and σ_k^2 are the mean and variance parameters for the k^{th} class

LDA for $p = 1$

- Assume also that $\sigma_1^2 = \dots = \sigma_K^2 = \sigma^2$: there is a **shared variance term** across all K classes, which for simplicity we can denote by σ^2 .
- Plugging **Eq. (2)** into **Eq. (1)**, we find that

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{j=1}^K \pi_j \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_j)^2\right)} \quad (3)$$

- The Bayes classifier involves **assigning an observation $X = x$ to the class for which Eq. (3) is the largest**

LDA for $p = 1$

- Taking the **logarithm of Eq. (3)** and rearranging the terms, we can show that the **classification of x** is equivalent to **assigning the observation** to the **class** for which

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k) \quad (4)$$

is the **largest**

- For instance, if $\mathbf{K} = 2$ and $\pi_1 = \pi_2$, then the **Bayes classifier** assigns an observation to class 1 if $2x(\mu_1 - \mu_2) > \mu_1^2 - \mu_2^2$, and to class 2 otherwise. In this case, the **Bayes decision boundary** corresponds to the point where

$$x = \frac{\mu_1^2 - \mu_2^2}{2(\mu_1 - \mu_2)} = \frac{\mu_1 + \mu_2}{2} \quad (5)$$

Overlapping between normal density functions

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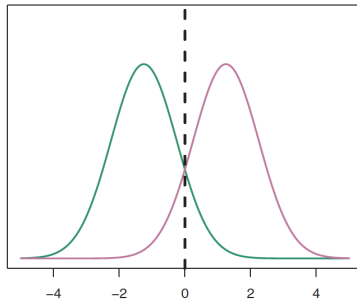
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The two normal density functions $f_1(x)$ ($\mu_1 = -1.25$, $\sigma^2 = 1$) and $f_2(x)$ ($\mu_2 = 1.25$, $\sigma^2 = 1$) represent **two distinct classes**. The two densities overlap, and so given that $X = x$, there is some **uncertainty** about the class to which the observation belongs.

Overlapping between normal density functions

- If we assume that an observation is **equally likely** to come from either class, i.e. $\pi_1 = \pi_2 = 0.5$, then by of Eq. (5), we see that the **Bayes classifier** assigns the observation to class 1 if $x < 0$ and class 2 otherwise
- We can compute the Bayes classifier because we know that X is drawn from a **Gaussian distribution** within each class, and we know all of the parameters involved
- In a **real-life situation**, we are **not able to calculate the Bayes classifier**: even if we are quite certain of our assumption that X is drawn from a Gaussian distribution within each class, we still have to estimate the **parameters** $\mu_1, \dots, \mu_K, \pi_1, \dots, \pi_K$, and σ^2

Linear Discriminant Analysis Algorithm

- **Linear Discriminant Analysis (LDA)** approximates the Bayes classifier by plugging estimates for π_k , μ_k and σ^2 into **Eq. (4)**. The estimates used are:

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i \quad (6)$$

$$\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 \quad (7)$$

where n is the total number of training observations, and n_k is the number of training observations in the k^{th} class

- The estimate for μ_k is simply the **average of all the training observations from the k^{th} class**, while σ^2 can be seen as a **weighted average of the sample variances for each of the K classes**

Discriminant functions

- Sometimes we have knowledge of the **class membership probabilities** $\pi_1, \dots, \pi_K$, which can be used directly. **In the absence** of this information, LDA estimates π_k using the **proportion of the training observations that belong to the k^{th} class**, namely

$$\hat{\pi}_k = \frac{n_k}{n}$$

- The **LDA classifier** finally assigns observation $X = x$ to the class for which

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

is the largest

- The **discriminant functions** $\hat{\delta}_k(x)$ are linear functions in x

Bayes vs LDA decision boundary

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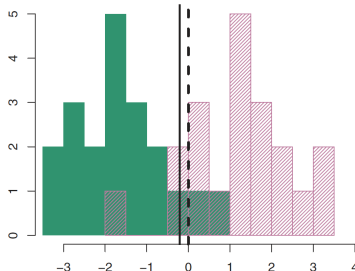
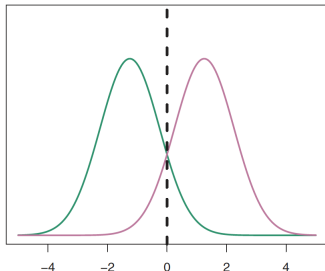
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20 observations, shown as histograms, are drawn for each of the **two classes**. The **dashed line** represents the **Bayes decision boundary** while the **solid line** represents the **LDA decision boundary**. We first estimate $\hat{\pi}_k$, $\hat{\mu}_k$, and $\hat{\sigma}^2$, then we compute the **LDA decision boundary** by assigning observations to the class k for which $\hat{\delta}_k$ is the largest.
Left: green class; Right: purple class

Decision Boundary

- Since $n_1 = n_2 = 20$, we have $\hat{\pi}_1 = \hat{\pi}_2$. **Decision boundary**: midpoint between the sample means for the two classes,

$$\frac{\hat{\mu}_1 + \hat{\mu}_2}{2}$$

- The **LDA decision boundary** is slightly to the left of the optimal Bayes decision boundary, which is

$$\frac{\mu_1 + \mu_2}{2} = 0$$

- **How well does the LDA classifier perform on this data?**
- **Bayes error rate**: 10.6%. **LDA test error rate**: 11.1%, i.e., 0.5% above the smallest possible error rate. Errors computed by generating **several test observations** since the dataset is simulated

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Multivariate Gaussian Distribution

- We now **extend** the LDA classifier to the case of **multiple predictors**. We will assume that $X = (X_1, X_2, \dots, X_p)$ is drawn from a **multivariate Gaussian distribution** with a class-specific **mean vector** and a common **covariance matrix**
- The multivariate Gaussian distribution assumes that each individual predictor follows a **one-dimensional** normal distribution, with some **correlation** between each pair of predictors.
- The **normal density of each predictor** takes the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right) \quad (8)$$

where μ_k and σ_k^2 are the mean and variance parameters for the k^{th} class

Examples of multivariate Gaussian distributions with $p = 2$

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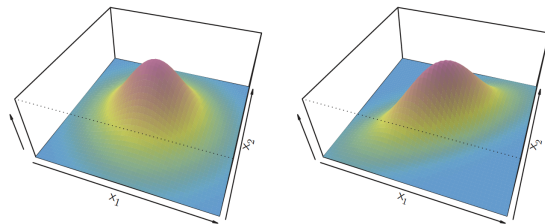
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The **height of the surface** at any particular point represents the **probability** that both X_1 and X_2 fall in a small **region around that point**. **Cut along the X_1 or X_2 axis:** cross-section with shape of a one-dimensional normal distribution.

Left: $Var(X_1) = Var(X_2)$ and $Cor(X_1, X_2) = 0$ (bell shape).

Right: correlated predictors or predictors with different variances (distorted bell shape)

Multivariate Gaussian density: notation

- To indicate that a p -dimensional random variable X has a **multivariate Gaussian distribution**, we write $X \sim N(\mu, \Sigma)$
- $E(X) = \mu$ is the **mean** of X (a vector with p components); $Cov(X) = \Sigma$ is the $p \times p$ **covariance matrix** of X
- The **multivariate Gaussian density** is defined as

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

Linear Discriminant Analysis for $p > 1$

- The LDA classifier assumes the **observations in the k^{th} class are drawn from a multivariate Gaussian distribution $N(\mu_k, \Sigma)$** , where μ_k is a class-specific **mean vector**, and Σ is a **covariance matrix** that is **common to all K classes**
- Plugging the density function for the k^{th} class, $f_k(X = x)$ reveals that **the Bayes classifier assigns an observation $X = x$ to the class for which**

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k \quad (9)$$

is the **largest**.

- This is the vector/matrix version of Eq. (4).

An example with 2 predictors and 3 classes

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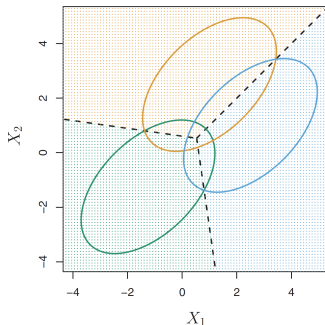
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Three equally-sized Gaussian classes are shown with **class-specific mean vectors** and a **common covariance matrix**. The three **ellipses** represent regions that contain 95% of the probability for each of the three classes



The **dashed lines** are the **Bayes decision boundaries**, representing the set of values x for which $\delta_k(x) = \delta_l(x)$ i.e.

$$x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k = x^T \Sigma^{-1} \mu_l - \frac{1}{2} \mu_l^T \Sigma^{-1} \mu_l$$

for $k \neq l$

An example with 2 predictors and 3 classes

- There are **three lines** representing the **Bayes decision boundaries** because there are three pairs of classes among the three classes. These three Bayes decision boundaries divide the predictor space into **three regions**.
- The Bayes classifier will classify an observation according to the region in which it is located.
- We need to **estimate the unknown parameters** $\mu_1, \dots, \mu_K, \pi_1, \dots, \pi_K$ and Σ .
- LDA **plugs** these estimates into Eq. 9 and **classifies** to the class for which $\hat{\delta}_k(x)$ is the **largest**.
- Notice: in Eq. 9 $\delta_k(x)$ is a **linear function** of x , i.e., the **LDA decision rule** depends on x only through a **linear combination** of its elements.

Bayes vs LDA Decision Boundaries

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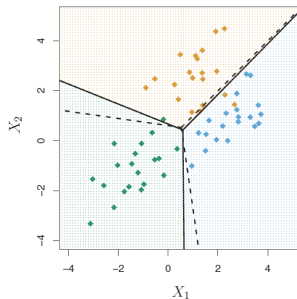
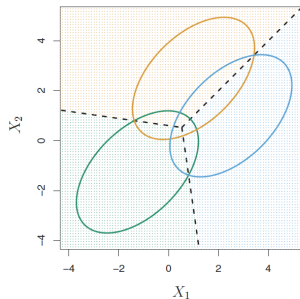
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20 observations drawn from each of the three classes are displayed, and the resulting **LDA decision boundaries** are shown as **solid black lines**. Overall, the LDA decision boundaries are pretty **close to the Bayes decision boundaries**, shown again as **dashed lines**. The **test error rates** for the Bayes and LDA classifiers are 0.0746 and 0.0770, respectively. This indicates that LDA is performing well on this data.

Default Example

- LDA on the **Default** dataset to predict whether or not an individual will default on the basis of credit card balance. The **LDA model fit** to the 10000 **training samples** results in a **training error rate of 2.75%**. This sounds like a **low** error rate, **but** a remark is needed.
- **Only 3.33%** of the individuals in the training sample **defaulted**. A simple but useless classifier that **always predicts** that each individual will not **default**, regardless of credit card balance, will result in an **error rate of 3.33%**.
- In other words, the trivial **null classifier** will achieve an **error rate** that is only a **bit higher than that of LDA**.
- In practice, a binary classifier such as this one can make **two types of error**: it can incorrectly assign an individual who **defaults** to the **no default** category, or it can do the **opposite**.

Confusion Matrix

- A **confusion matrix** is a convenient way to display **which of these two types of errors are made**

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9,644	252	9,896
	Yes	23	81	104
Total		9,667	333	10,000

- LDA predicted that a total of **104** people would **default**. Of these people, **81** actually defaulted and **23** did not.
- Hence, **only 23 out of 9667 of the individuals who did not default were incorrectly labeled**. This looks like a pretty **low error rate**!
- But, **of the 333 individuals who defaulted, 252 (or 75.7%) were missed by LDA**. So while the overall error rate is low, the **error rate among individuals who defaulted is very high**.

Class-specific performance

- **Sensitivity:** percentage of **true defaulters** that are **identified** (here, 24.3%).
- **Specificity:** percentage of **non-defaulters** that are **correctly identified**, here (99.8%).
- **Why** does LDA do such a **poor** job of classifying the customers who **default**? Why does it have such a **low sensitivity**? LDA is trying to approximate the Bayes classifier, which has the lowest total error rate out of all classifiers. That is, the Bayes classifier will yield the **smallest possible total number of misclassified observations**, *irrespective of which class the errors come from*.
- In contrast, a **credit card company** might particularly wish to **avoid incorrectly classifying** an individual who will **default**.
- It is possible to **modify LDA** in order to develop a classifier that **better meets the credit card company's needs**.

Threshold

- The **Bayes classifier** works by assigning an observation to the class for which $p_k(X)$ is the **greatest**. With $K = 2$, we assign an observation to the default class if

$$Pr(\text{default} = \text{yes} \mid X = x) > 0.5$$

- Thus, the **Bayes classifier**, and by extension **LDA**, uses a **threshold** of **50%** for the posterior probability of default in order to assign an observation to the **default class**.
- To modify our predictions and **increase the sensitivity**, we can consider **lowering the threshold**. For instance, we might label any customer - with a posterior probability of default above **20%** - to the default class

$$Pr(\text{default} = \text{yes} \mid X = x) > 0.2$$

Confusion Matrix with the new threshold

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		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9,432	138	9,570
	Yes	235	195	430
Total		9,667	333	10,000

Now LDA predicts that **430** individuals will **default**. Of the 333 individuals who default, LDA **correctly predicts all but 138, or 41.4%**. This is a **strong improvement** over the error rate on true defaulters of 75.7% that resulted from using the threshold of 50%. However, there is a **cost**: now **235 individuals who do not default are incorrectly classified**. As a result, the **overall error rate has increased slightly to 3.73%**.

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Trade-off between Errors and Threshold

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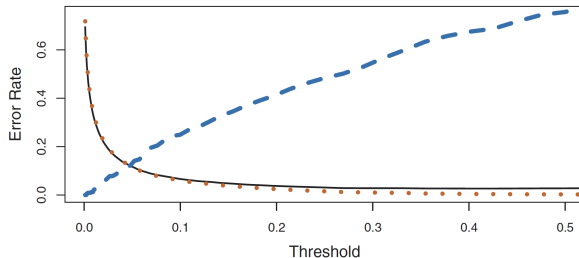
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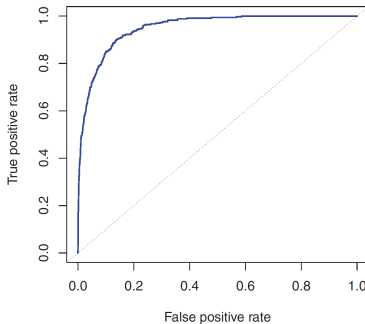


Various **error rates** are shown as a **function of the threshold value**. Using a threshold of **0.5** minimizes the **overall error rate (black solid line)** (Bayes classifier). But with threshold 0.5 the **error rate among the individuals who default** is quite **high (blue dashed line)**. As the **threshold is reduced**, the error rate among individuals who **default decreases** steadily, but the error rate among the individuals who **do not default increases**.

Threshold calibration and the ROC curve

- Threshold calibration must be based on **domain knowledge** (e.g., information about the **costs associated with default**).
- The *ROC curve* is a popular chart for **simultaneously displaying two types of errors for all possible thresholds**.
- The name “ROC” is historic (**Receiver Operating Characteristics**, communication theory)
- The **overall performance of a classifier**, summarized over all possible thresholds, is given by the **Area Under the ROC Curve (AUC)**

Example of ROC curve for Default dataset



ROC curve for the LDA classifier on the **Default** data. It traces the **two errors (True Positive and False Positive)** varying the threshold. True positive rate is **sensitivity** and the False positive rate is **1-specificity**. **AUC** is 0.95. The **dotted line** represents the ROC for the "no information" classifier (AUC is 0.5, and would be the result **if predictors were not co-related with response**).

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Important measures for classification and diagnostic testing